

ResQ – Emergency Response Mobile Application

1. Problem Statement

In many schools and campus environments, emergencies such as theft, medical incidents, harassment, and security threats occur unexpectedly and require immediate response. Existing emergency systems are often slow, centralized, or inaccessible, relying heavily on administrators or external authorities who may not be immediately available. Victims are left vulnerable during the most critical moments, with no guaranteed real-time assistance. There is a lack of a smart, community-driven emergency response system that combines instant alerts, proximity-based help, and trusted peer support within a controlled environment such as a school campus.

2. Project Overview

ResQ is a campus-focused emergency response mobile application designed to provide fast, reliable, and layered assistance during emergencies. The system connects residents (students and staff), administrators, emergency contacts, and verified nearby helpers to ensure that help is always available, even when one response layer fails.

3. Project Objectives

- To provide instant emergency reporting with real-time location sharing.
- To alert nearby users for faster response during incidents like theft or harassment.
- To notify trusted emergency contacts automatically.
- To enable administrators to monitor and manage emergencies effectively.
- To introduce a verified peer-assistance system for sensitive and immediate cases.
- To create a scalable model suitable for campus environments.

4. Core Features

4.1 Emergency Alert System

Users can trigger an emergency alert which sends their location and incident type to administrators and nearby users.

4.2 Proximity-Based Alerts

Users within a defined radius are notified to provide immediate assistance, especially useful for theft and urgent situations.

4.3 Emergency Contact Notification

If admin response is delayed, the system automatically contacts pre-selected trusted individuals with live location details.

4.4 Verified Peer Assistance

Verified helpers (e.g., females or trained volunteers) can respond to sensitive emergencies when official help is unavailable. This ensures safety, inclusivity, and rapid intervention.

4.5 Admin Dashboard

Administrators can view emergencies on a map, respond, manage users, and track incident history.

4.6 Map & History

- Map view showing active emergencies.
- History tab for tracking resolved incidents and analytics.

5. Justification of Features

Each feature is designed to solve a real-world delay or failure in emergency response. By introducing multiple fallback layers—nearby users, emergency contacts, and verified peers—ResQ ensures that help is never dependent on a single point of failure.

6. Scope Limitation

For this project, ResQ is limited to school or campus environments to ensure controlled user verification, manageable data scope, and realistic implementation within a final-year academic setting.

7. Expected Outcomes

- Faster emergency response times.
- Improved safety awareness on campus.
- Reduced dependence on centralized authorities alone.
- A scalable emergency response framework for future expansion.

8. Future Enhancements

- Integration with SMS and emergency services.
- AI-based emergency severity detection.
- City-wide deployment and smart surveillance integration.
- Wearable device support.